

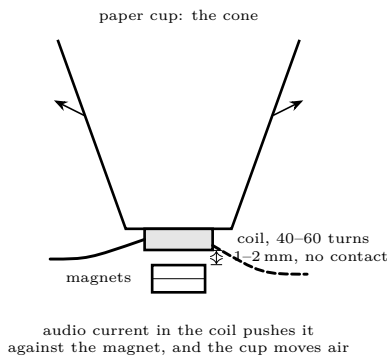
Demo: Build a Speaker

Turn a coil, a magnet and a paper cup into music

The same idea as the motor, running fast enough to hear

Lesson Demo D · 30 minutes · after Lesson 4 — audio output only, safe

What you need	
• Magnet wire, 28–32 AWG, about 10 m • A paper or plastic cup, or a stiff card disc • Two or three neodymium disc magnets • A 3.5 mm audio cable you are willing to cut, or a head-	• phone socket breakout • A phone or small amplifier as the source • Tape, sandpaper, a round former (a AAA battery or a marker pen)



Predict first	
A speaker has no moving parts you can see and no valves. Music is a pattern of air pressure. What has to happen, physically, to turn a wire carrying a changing current into a pattern of air pressure? Write it in one sentence before you build.	
Coil in a field	Exactly Demo A and Lesson 4: current in a magnetic field feels a force. Here the current is the audio signal, changing thousands of times a second.
Push and pull	Because audio current alternates, the force alternates. The coil is pushed away and pulled back, hundreds or thousands of times a second.
Moving air	The coil is glued to a cone. The cone moves air. Moving air is sound. That is the entire chain, and there is nothing else in a loudspeaker.
It works backwards too	Speak into it and the cone moves the coil in the field, inducing a small voltage — Lesson 5. A speaker is a microphone run in reverse, and you can prove it with the multimeter.

Build it

1. **Wind the coil** around a AAA battery or a marker pen: 40–60 turns, all the same direction, leaving 20 cm tails. Slide it off and tape it so it holds its ring shape.
2. **Sand the enamel** off both tails until bright copper shows. This is the step that gets skipped and it is why it will not work.
3. **Tape the coil flat** to the bottom of the cup, centred.
4. **Put the magnets under it**, close but not touching — 1 to 2 mm. The coil must be free to move.
5. **Connect the tails** to the audio cable’s signal and ground conductors, and play something loud with a lot of bass.
6. Hold the cup lightly. You will feel it before you hear it.

Break it

More turns	Louder — more current-carrying wire in the field. Also more resistance. Find where it stops helping.
Move the magnet	Further away is quieter, and the fall-off is steep. Same distance-dependence as the eddy-current pendulum. <i>see the Demo: Eddy Currents sheet</i>
Change the cone	A big stiff card is louder and better at low notes; a small cup is quieter and thinner. This is the whole of loudspeaker design in one experiment.
Run it backwards	Disconnect the audio, connect the multimeter on millivolts AC, and shout into the cup. You are looking at your own voice as a voltage.
Volume	Keep it modest. An unloaded coil is close to a short circuit and a phone output will not enjoy it.

The point

Every loudspeaker in the house — phone, television, car, headphones — is this object, made precisely. Once a teenager has built one out of a cup, the sentence “a speaker is a coil in a magnetic field glued to a piece of paper” stops being a claim and becomes something they have held in their hand.

This lesson is low voltage. Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

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